17 The real distance between two towns is 75 km.
On a map, the distance between the two towns is 3 cm.

(a) Complete the following statement.

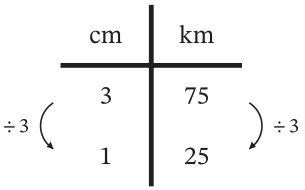
1 cm on the map represents a real distance of ..... km.

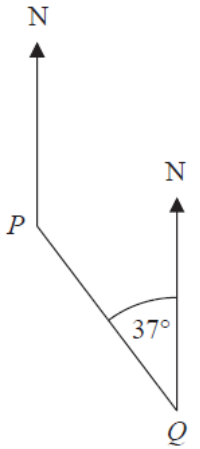
75 ÷ 3 = 25 km per cm

1 mark
1 cm means 25 km.

(1)

The diagram shows the position of two castles, P and Q.





1 cm represents 25 km.

Mike has to work out the bearing of P from Q.
Here is Mike's working.

$$180 - 37 = 143$$
The bearing of P from $Q = 143°$

Mike's answer is wrong.

(b) What mistake has Mike made?

He subtracted from 180°, but should subtract from 360°.

Bearing at Q is clockwise from north: 360° – 37° = 323°.

1 mark
Explain his use of 180 instead of 360.

(1)

(Total for Question 17 is 2 marks)